

AASD 4016

Full Stack Data Science Systems

Applied AI Solutions Developer Program



Module 6

ML Canvas

Vejey Gandyer



Agenda



ML Canvas

Why

Steps

Examples



ML Canvas

The Machine Learning Canvas

Designed for:

Designed by:

Date:

Iteration:

Decisions  How are predictions used to make decisions that provide the proposed value to the end-user?	ML task  Input, output to predict, type of problem.	Value Propositions  What are we trying to do for the end-user(s) of the predictive system? What objectives are we serving?	Data Sources  Which raw data sources can we use (internal and external)?	Collecting Data  How do we get new data to learn from (inputs and outputs)?
Making Predictions  When do we make predictions on new inputs? How long do we have to featurize a new input and make a prediction?	Offline Evaluation  Methods and metrics to evaluate the system before deployment.		Features  Input representations extracted from raw data sources.	Building Models  When do we create/update models with new training data? How long do we have to featurize training inputs and create a model?
	Live Evaluation and Monitoring  Methods and metrics to evaluate the system after deployment, and to quantify value creation.			

THE MACHINE LEARNING CANVAS

Designed for:

Designed by:

Date:

Iteration:

PREDICTION TASK  Type of task? Entity on which predictions are made? Possible outcomes? Wait time before observation?	DECISIONS  How are predictions turned into proposed value for the end-user? Mention parameters of the process / application that does that.	VALUE PROPOSITION  Who is the end-user? What are their objectives? How will they benefit from the ML system? Mention workflow/interfaces.	DATA COLLECTION  Strategy for initial train set & continuous update. Mention collection rate, holdout on production entities, cost/constraints to observe outcomes.	DATA SOURCES  Where can we get (raw) information on entities and observed outcomes? Mention database tables, API methods, websites to scrape, etc.
IMPACT SIMULATION  Can models be deployed? Which test data to assess performance? Cost/gain values for (in)correct decisions? <u>Fairness constraint</u>?	MAKING PREDICTIONS  When do we make real-time / batch pred.? Time available for this + featurization + post-processing? Compute target?		BUILDING MODELS  How many prod models are needed? When would we update? Time available for this (including featurization and analysis)?	FEATURES  Input representations available at prediction time, extracted from raw data sources.
MONITORING  Metrics to quantify value creation and measure the ML system's impact in production (on end-users and business)?				



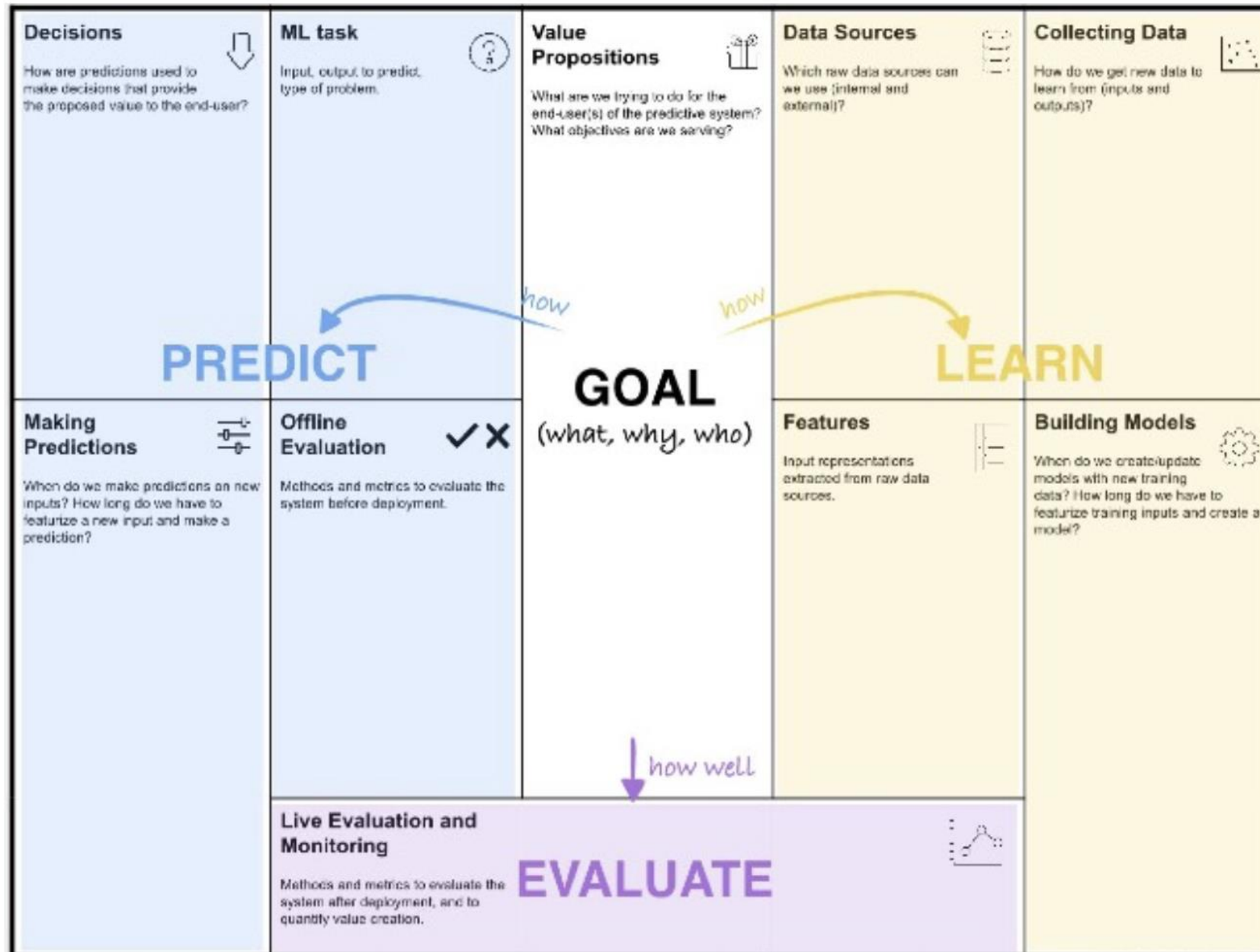
ML Canvas

Learn

Predict

Evaluate





THE MACHINE LEARNING CANVAS

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Further Reading

ML Canvas

<https://www.ownml.co/machine-learning-canvas>